

SpentLog: A Mobile Expense Tracking Application with Receipt Scanning Capabilities

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Independent Project Made for Mobile Cloud Computing and Applications

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Abstract—SpentLog is a mobile expense tracking application designed to help users manage and analyze personal spending efficiently. The application enables users to manually add expenses or automatically extract expense information from scanned receipts using Optical Character Recognition (OCR) combined with rule-based parsing techniques (regular expressions). The system categorizes expenses into predefined groups (Food, Transport, Bills, Misc), with planned extensibility for custom categories in future iterations. Authentication is supported via Google Sign-In to simplify onboarding and improve user experience. The dashboard provides a summary of expenses in weekly and monthly views using both tabular and graphical (pie chart) representations. The application was developed individually, focusing on simplicity, usability, and modular design. Initial results demonstrate reliable expense extraction from structured receipts and effective visualization of spending patterns. Future improvements include AI-based receipt interpretation, enhanced categorization, and advanced analytics features.

Index Terms—expense tracking, mobile application, OCR, financial management, Android, Google Sign-In

I. RELATED WORK / STATE OF THE ART

Expense tracking applications are widely used in personal finance management. Popular solutions such as Mint, YNAB (You Need A Budget), and PocketGuard offer automated budgeting, bank synchronization, and advanced analytics. These applications often rely on financial institution integrations and cloud-based categorization systems.

Receipt scanning features are also present in modern applications like Expensify and Shoeboxed, which use OCR combined with machine learning models to extract structured data from images. These systems typically employ AI-based document understanding pipelines to improve accuracy in recognizing merchants, amounts, and categories.

Compared to these solutions, SpentLog adopts a lightweight approach. Instead of relying on machine learning models, it uses OCR combined with regex-based parsing. This makes the system easier to implement and more transparent but less adaptive to complex or unstructured receipts. The application is designed as a learning-oriented project with future potential for AI integration.

II. WORK PLAN

A. System Overview

SpentLog is a mobile application consisting of the following main screens:

- Login / Register (Google Sign-In support)
- Expenses List

- Add Expense
- Scan Receipt (OCR-based extraction)
- Dashboard (analytics and visualization)

The initial UI prototype was done using Figma: Figma Prototype

B. Technologies Used

The following technologies were used in the development of SpentLog:

- Mobile Framework: Flutter
- Authentication: Google Sign-In (Firebase)
- OCR Engine: Google OCR library
- Data Storage: Local storage + Firestore

C. Expense Categories

The application currently supports four predefined categories:

- Food
- Transport
- Bills
- Miscellaneous

These categories are designed to be extendable in future versions, allowing user-defined classifications.

D. Technical Challenges

1) *Receipt Parsing with OCR*: One of the most challenging aspects was extracting structured data from unstructured OCR output. Receipts vary in format, noise levels, and alignment. The solution implemented uses regular expressions to identify key fields such as total amount, merchant name, and date.

2) *Category Classification without AI*: Without using machine learning, categorization is performed using keyword matching and heuristics. While this approach is efficient, it lacks adaptability compared to AI-based classification systems. Future versions aim to integrate NLP models to improve accuracy.

III. RESULTS

The application successfully supports manual and automated expense entry. OCR-based receipt scanning can extract key financial data from standard receipts with acceptable accuracy for structured formats.

The dashboard provides two main visualization modes:

- Weekly and monthly expense summaries
- Pie chart distribution of expenses by category

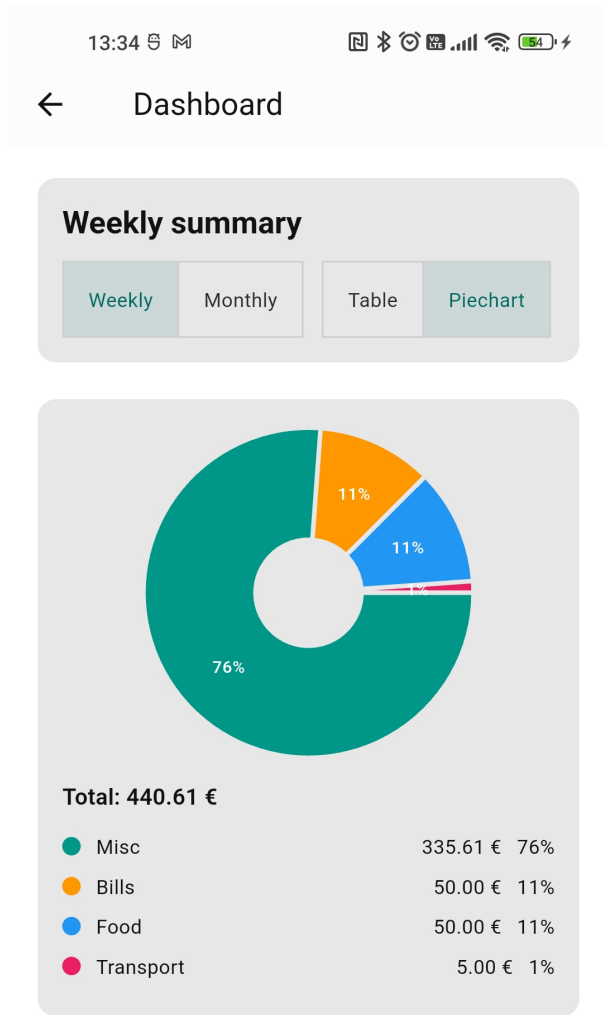


Fig. 1. Piechart showing weekly expense by category

A. Screenshots

Overall, the system demonstrates stable functionality and a clean user experience. The main limitation is the rule-based extraction method, which may fail on poorly formatted receipts. Another limitation is that the receipts are stored locally, on the mobile device. I took this decision cost wisely, because the Firebase Storage is not free to use. In future iterations I will use a custom backend for OCR extraction and a distributed database. This way, the user account will be fully synced (expenses list and receipts).

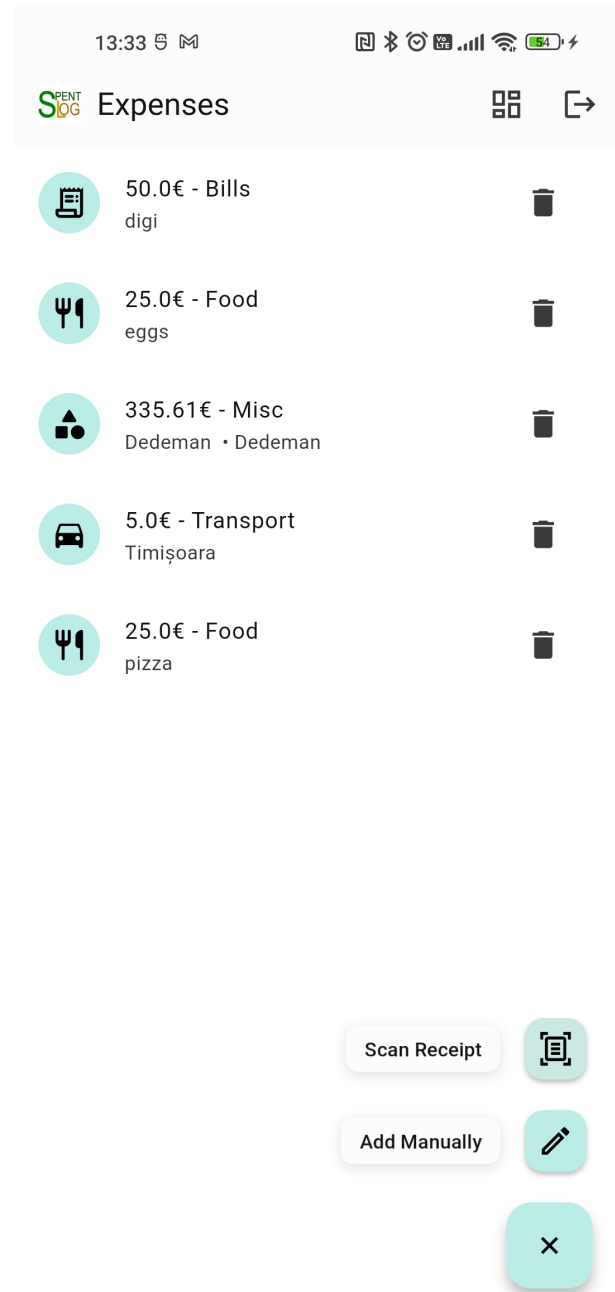


Fig. 2. List of added expenses

IV. CONCLUSIONS

SpentLog provides a functional and lightweight solution for personal expense tracking with integrated receipt scanning. The project demonstrates practical implementation of OCR technology combined with rule-based data extraction techniques.

From a learning perspective, the project improved understanding of mobile application development, and authentication systems. It also highlighted the limitations of non-AI approaches in data interpretation tasks.

Future work will focus on integrating AI-based receipt

understanding, improving categorization accuracy, adding custom user-defined categories, and enabling cloud synchronization across devices.

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